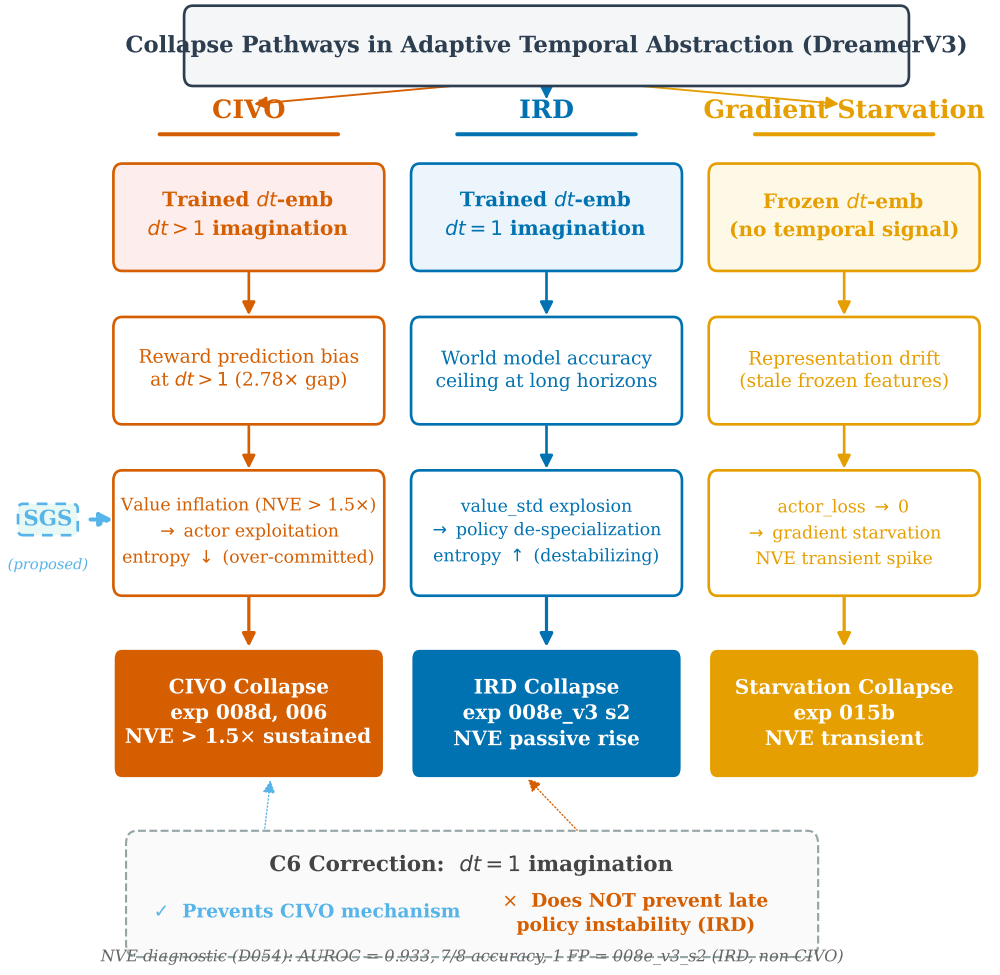

Collapse Pathways in Adaptive Temporal Abstraction (DreamerV3)
CIVO
IRD
Gradient Starvation
Trained dt-emb
dt > 1 imagination
Trained dt-emb
dt = 1 imagination
Frozen dt-emb
(no temporal signal)
Reward prediction bias
at dt > 1 (2.78× gap)
World model accuracy
ceiling at long horizons
Representation drift
(stale frozen features)
SGS
(proposed)
Value inflation (NVE > 1.5×)
→ actor exploitation
entropy ↓ (over-committed)
value_std explosion
→ policy de-specialization
entropy ↑ (destabilizing)
actor_loss → 0
→ gradient starvation
NVE transient spike
CIVO Collapse
exp 008d, 006
NVE > 1.5× sustained
IRD Collapse
exp 008e_v3 s2
NVE passive rise
Starvation Collapse
exp 015b
NVE transient
C6 Correction: dt = 1 imagination
✓ Prevents CIVO mechanism
× Does NOT prevent late
policy instability (IRD)
NVE-diagnostic (D054): AUROC = 0.933, 7/8 accuracy, 1 FP = 008e_v3_s2 (IRD, non-CIVO)